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Studierichting: Informatica
Oefeningenreeks 7B nr. 7.1

$f : \mathbb{R}^2 \to \mathbb{R}^2 : (x, y) \to (x^2 - y^2, 2xy)$

$Df = \begin{pmatrix} 2x & 2y \\ -2y & 2x \end{pmatrix}$

$\det(Df) = (2x)(2x) - (-2y)(2y) = 4x^2 + 4y^2$

$Df^{-1} = \frac{1}{4x^2 + 4y^2} \begin{pmatrix} 2x & 2y \\ -2y & 2x \end{pmatrix}$

$$Df^{-1} = \begin{pmatrix} \frac{2x}{4x^2 + 4y^2} & \frac{2y}{4x^2 + 4y^2} \\ \frac{-2y}{4x^2 + 4y^2} & \frac{2x}{4x^2 + 4y^2} \end{pmatrix}$$

References